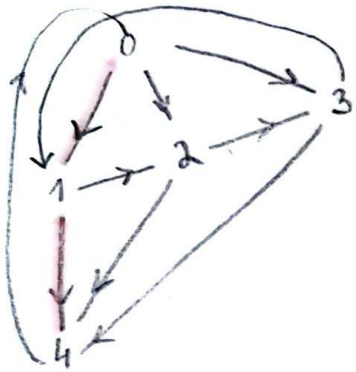


Lab 2 Lowest length path between s and t using backward BFS from t



Nin dictionary

0: [4]
1: [0, 3]
2: [0, 1]
3: [0, 2]
4: [1, 2, 3]

$s=0$ $t=4$

initialization

iteration 1

1.1

1.2

1.3

iteration 2

2.1.

2.2.

iteration 3

iteration 4

iteration 5

current x y der. neighb.

x

y

queue

visited

dist

parent

-

-

$\leftarrow 4 \leftarrow$

0 1 2 3 4
F F F F F

0 1 2 3 4
0

0 1 2 3 4
-1 -1 -1 -1 -1

4

1

$\leftarrow 1 \leftarrow$

0 1 2 3 4
F T F F F

0 1 2 3 4
1 0

0 1 2 3 4
-1 4 -1 -1 -1

2

$\leftarrow 1 2 \leftarrow$

0 1 2 3 4
F T T F F

0 1 2 3 4
1 1 0

0 1 2 3 4
-1 4 4 -1 -1

3

$\leftarrow 1 2 3 \leftarrow$

0 1 2 3 4
F T T T F

0 1 2 3 4
1 1 1 0

0 1 2 3 4
-1 4 4 4 -1

1

-

$\leftarrow 2 3 \leftarrow$

0 1 2 3 4
F T T T T

0 1 2 3 4
1 1 1 1

0 1 2 3 4
1 4 4 4 -1

0

$\leftarrow 2 3 0 \leftarrow$

0 1 2 3 4
T T T T T

2 1 1 1 1

- " -

3

$\leftarrow 2 3 0 \leftarrow$

0 1 2 3 4
T T T T T

skip

- " -

2

$\leftarrow 3 0 \leftarrow$

0 1 2 3 4
T T T T T

- " -

- " -

3

$\leftarrow 1 0 \leftarrow$

0 1 2 3 4
T T T T T

- " -

- " -

0

$\leftarrow \leftarrow$

- " -

stop

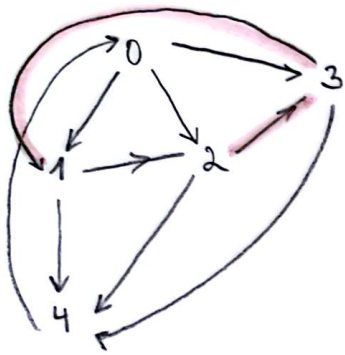
the path is built from parent dict

0	1	2	3	4
1	4	4	4	

 beginning with $s=0$

$s=0 \rightarrow \text{parent}[0]=1 \rightarrow \text{parent}[1]=4=t \Rightarrow \text{path} = 0 \rightarrow 1 \rightarrow 4$

$\text{len} = \text{dist}[s] = \text{dist}[0] = 2$



Nin dictionary

0: [4]

1: [0, 3]

2: [0, 1]

3: [0, 2]

4: [1, 2, 3]

$s = 2$ $t = 1$

initialization

iteration 1

1.1

1.2.

iteration 2

2.1

iteration 3

iteration 4

iteration 5

x

y

queue

visited

dist.

parent

← 1 ←

0 1 2 3 4
F F F F F

0 1 2 3 4
0

0 1 2 3 4
-1 -1 -1 -1 -1

0 1 2 3 4
F T F F F

0 1 2 3 4
0

0 1 2 3 4
-1 -1 -1 -1 -1

← 0 ←

0 1 2 3 4
T T F F F

0 1 2 3 4
1 0

0 1 2 3 4
1 -1 -1 -1 -1

← 0 3 ←

0 1 2 3 4
T T F T F

0 1 2 3 4
1 0 1

0 1 2 3 4
1 -1 -1 1 -1

← 3 ←

0 1 2 3 4
T T F T F

0 1 2 3 4
1 0 1

0 1 2 3 4
1 -1 -1 1 -1

← 3 4 ←

0 1 2 3 4
T T F T T

0 1 2 3 4
1 0 1 2

0 1 2 3 4
1 -1 -1 1 0

← 4 ←

0 1 2 3 4
T T F T T

0 1 2 3 4
1 0 1 2

0 1 2 3 4
1 -1 -1 1 0

← 4 ←

0 1 2 3 4
T T F T T

skip

- - -

← 4 2 ←

0 1 2 3 4
T T T T T

0 1 2 3 4
1 0 2 1 2

0 1 2 3 4
1 0 3 1 0

← 2 ←

- - -

- - -

- - -

- - -

- - -

stop

the path is built from parent dict.

0	1	2	3	4
1	0	3	1	0

 beginning with $s = 2$

$s = 2 \rightarrow \text{parent}[2] = 3 \rightarrow \text{parent}[3] = 1 = t \Rightarrow \text{path} = 2 \rightarrow 3 \rightarrow 1$

$\text{len} = \text{dist}[s] = \text{dist}[2] = 2.$

Graph with 1k vertices

```
16 lowest length path between 2 vertices : backwards BFS
0 exit
> 16
start: 1
end: 100
[1, 104, 445, 856, 946, 558, 100]
length: 6
```

```
16 lowest length path between 2 vertices : backwards BFS
0 exit
> 16
start: 100
end: 1
[100, 782, 335, 301, 340, 1]
length: 5
```

Graph with 10k vertices

```
> 16
start: 1
end: 100
[1, 8493, 2013, 3797, 914, 8515, 7530, 9126, 100]
length: 8

16 lowest length path between 2 vertices : backwards BFS
0 exit
> 16
start: 100
end: 1
[100, 328, 9684, 2367, 4936, 2133, 4542, 1]
length: 7
```

Graph with 100k vertices

```
16 lowest length path between 2 vertices : backwards BFS
0 exit
> 16
start: 1
end: 100
[1, 43594, 46357, 75840, 2795, 6232, 68946, 86274, 89221, 100]
length: 9
```

```
16 lowest length path between 2 vertices : backwards BFS
0 exit
> 16
start: 100
end: 1
[100, 25943, 99350, 47899, 96479, 65007, 19499, 21231, 1]
length: 8
```